WHAT THE CELL · DP BIOLOGY · WORKSHEET · SL STATEMENTS C1.2.1–6

C1.2 Cell respiration

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

8 MARKS

1 STATE the full name of ATP. [C1.2.1] [1]

2 LIST three processes in a cell that use energy from ATP. [C1.2.2] [3]

3 WRITE the word equation for aerobic respiration, with glucose as the substrate. [C1.2.5] [2]

4 STATE the waste product of anaerobic respiration in humans. [C1.2.5] [1]

5 STATE the part of the cell in which anaerobic respiration occurs. [C1.2.5] [1]

Section B: Understanding

12 MARKS

6 EXPLAIN why ATP is suited to being the energy currency of a cell. [C1.2.1] [3]

7 DESCRIBE the energy changes when ATP and ADP interconvert. [C1.2.3] [3]

8 DISTINGUISH between cell respiration and gas exchange. [C1.2.4] [3]

9 **EXPLAIN** why humans use anaerobic respiration in a short sprint, even though the yield of ATP is small. [3]

[C1.2.5]

Section C: Working with data

14 MARKS

All data in this section are invented for practice. The patterns are realistic.

10 [C1.2.6]

A respirometer contained 5 g of germinating seeds and a CO_2 absorber. The capillary tube has a cross-section of 1 mm².

TEMPERATURE / °C	10	20	30	40
DISTANCE MOVED IN 10 MIN / MM	8	16	34	30

a **CALCULATE** the rate of respiration at 30°C, in mm³ per minute per gram. [3]

b **DESCRIBE** the trend. [2]

c **EXPLAIN** the change between 30°C and 40°C. [2]

d **EXPLAIN** why the drop moves towards the seeds. [3]

e **OUTLINE** a suitable control, and why it is needed. [2]

f **STATE** two variables that must be controlled. [2]

Section D: Longer answer

6 MARKS

11 **COMPARE AND CONTRAST** the two types of respiration in humans. [C1.2.5] [6]